UCT Mathematics Society presents

UCT Integration Bee

in collaboration with
UCT Department of Mathematics and Applied Mathematics

Qualifier Test
MEMORANDUM
11/12 September 2024

"Anyone can differentiate. Integration is an artform."

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(1) [1 *mark*] $\int \sec^2 x - \frac{\sec^2 x - \tan^2 x - \cos^2 x}{\csc^2 x - \cot^2 x - \sin^2 x}\ dx$

Answer:

$$x \checkmark$$

(2) [1 *mark*] $\int x(1+x^2)^{2/3}\ dx$

Answer:

$$\frac{3(x^2+1)^{5/3}}{10} \checkmark$$

(3) [1 *mark*] $\int \frac{1}{\sqrt{25-64x^2}}\ dx$

Answer

$$\frac{1}{8}\arcsin\left(\frac{8x}{5}\right) \checkmark$$

(4) [2 *marks*] $\int \frac{\cos x}{(1+\sin^2 x)\arctan(\sin x)}\ dx$

Answer:

$$\ln(\arctan(\sin x)) \checkmark\checkmark$$

(5) [2 *marks*] $\int x^2 \ln x\ dx$

Answer:

$$\frac{x^3}{9}(3\ln x - 1) \checkmark\checkmark$$

(6) [2 *marks*] $\int \frac{57^{\sqrt{x}}}{\sqrt{x}}\ dx$

Answer:

$$\frac{2}{\ln 57}57^{\sqrt{x}} \checkmark\checkmark$$

(7) [2 *marks*] $\int \sin^3 x\ dx$

Answer:

$$\frac{\cos^3 x}{3} - \cos x \checkmark\checkmark$$

(8) [3 *marks*] $\int x^2 \sin x \cos x\ dx$

Answer:

$$\frac{1}{8}\Big(2x\sin(2x) + (1-2x^2)\cos(2x)\Big)\checkmark\checkmark\ \checkmark$$

(9) [3 *marks*] $\int \sin^3(x)\cos^3(x)\ dx$

Answer:

$$\frac{\sin^4(x)}{4} - \frac{\sin^6(x)}{6}\checkmark\checkmark\ \checkmark$$

(10) [3 *marks*] $\int \frac{2x+3}{(x-1)(x^2+1)}\ dx$

Answer:

$$\frac{5}{2}\ln|x-1| - \frac{1}{2}\arctan x - \frac{5}{4}\ln(x^2+1)\checkmark\checkmark\ \checkmark$$

(11) [3 *marks*] $\int \frac{1}{x^2+2x}\ dx$

Answer:

$$\frac{1}{2}\ln\left|\frac{x}{x+2}\right|\checkmark\checkmark\ \checkmark$$

(12) [4 *marks*] $\int \frac{1}{\sqrt{1-x^2}\arcsin x - \arcsin x - x^2}\ dx$

Answer:

$$2\sqrt{1+\arcsin x}\checkmark\checkmark\ \checkmark\checkmark$$

(13) [4 *marks*] $\int \frac{1+2x^{2023}}{x+x^{2024}}\ dx$

Answer:

$$\ln|x| + \frac{1}{2023}\ln(x^{2023}+1)\checkmark\checkmark\ \checkmark\checkmark$$

(14) [5 *marks*] $\int e^x \sin x \cos x\ dx$

Answer:

$$\frac{e^x}{10}\Big(\sin(2x) - 2\cos(2x)\Big)\checkmark\checkmark\ \checkmark\checkmark\ \checkmark$$

(15) [5 *marks*] $\int \sqrt{1+e^x}\ dx$

Answer:

$$\ln\Big(\frac{\sqrt{e^x+1}+1}{\sqrt{e^x+1}-1}\Big) + 2\sqrt{e^x+1}\checkmark\checkmark\ \checkmark\checkmark\ \checkmark$$

(16) [6 *marks*] $\int \frac{(1-x^2)^{2/3}}{x^6}\ dx$

There was a typo in this question which made it impossible to find a closed form solution. For those who are interested, the intended integral $\int \frac{(1-x^2)^{3/2}}{x^6}\ dx$ has the following solution:

$$-\frac{\sqrt{1-x^2}(x^2-1)^2}{5x^5}\checkmark\checkmark\ \checkmark\checkmark\ \checkmark\checkmark$$

(17) [6 *marks*] $\int \frac{1}{x^3-1}\ dx$

Answer:

$$-\frac{1}{6}\Big(\ln(x^2+x+1) + 2\sqrt{3}\arctan\big(\frac{2x+1}{\sqrt{3}}\big) - 2\ln|2x+1|\Big)\checkmark\checkmark\ \checkmark\checkmark\ \checkmark\checkmark$$

(18) [7 *marks*] $\int x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x + \cdots}}}\, dx$

Answer:

$$\frac{1}{4}\left(x^2 + x\sqrt{x^2+4}\right) + \ln|\sqrt{x^2+4} + x| \checkmark\checkmark\ \checkmark\checkmark\ \checkmark\checkmark\ \checkmark$$